Comp.

College of Engineering, Pune

S.Y.B.Tech - Computer Engineering and Information Technology

End Semester Exam

CT-207 – MICROPROCESSOR TECHNIQUES

Date-27/04/2013 Timing: 3 hrs.

Academic Year: 2012- 13 Max. Marks: 100

Instructions:

1. Answer all Questions.
2. Draw neat diagrams wherever necessary.
3. Make appropriate assumptions if required.
4. Figures to the right indicate full marks.

			Marks
Q. 1.		Explain with suitable schematic for interfacing a 16 KWORD RAM, using 16 KB RAM chips, an 8 KWORD EPROM using 8 KB EPROM chips to 8086 microprocessor and an 8255 with port A address 8000H.The starting address of the RAM is 00000H and the last address of the EPROM is FFFFFH. Give the address map for memory and I/O.	18
Q. 2.	A.	What addressing mode is used the following instruction a. MOV AX, BX b. MOV AX, [DI] c. MOV AX,[BX+SI+04] d. MOV CX, 2342H	04
	B.	Give the significance of O Flag and T Flag of 8086.	06
	C.	Explain following instructions with examples a. PUSHF b. REPZ	06
Q.3.	A.	What is the difference between maskable and non-maskable interrupts?	04
	B.	Write an assembly program to calculate the factorial of a number N.	12
	OR		
	B.	Give the interfacing of 16 keys key matrix using 8255 in 8086 based system. Write an assembly program to read the key number of key pressed.	12

Q.4 **A.** What are the contents of BUFF2 after the execution of following 8086 assembly language program? **08**

```
.model small
.data
    BUFF1  db "IMAGE"
    BUFF2  db 5 DUP (?)
.code
           MOV AX, @data
           MOV DS, AX
           MOV ES, AX
           STD
           MOV CX, 0005H
           LEA SI,  BUFF1+4
           LEA DI, BUFF2
NEXT:      MOVSB
           ADD DI,0002H
           LOOP    NEXT
           END
```

OR

A. Write an ALP in 8086 to display the string "COEP" on display terminal. **08**

B. What is EOI command in 8259 Programmable Interrupt Controller? **05**

C. What are the possible errors getting detected and recorded in serial communication with the use of 8251 USART? **05**

Q.5 **A.** What are the different modes of operation of 8253/8254 Programmable interval timer? Explain mode 2 & mode 4 with waveforms. **07**

B. Explain the following terms w.r.t 8279 Programmable Keyboard / Display Interface **09**

1. Two key lock out
2. N Key roll over
3. Scanned sensor matrix mode

Q.6 **A.** Explain demand transfer mode and block transfer mode of 8237 DMA controller. **06**

B. Explain the interfacing and initialization of 8259 PIC with 8086 for managing 8 interrupt requests in default priority mode. **10**